Farhan Chatterjee

FPGA Design Engineer

RESUME_1111

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Total Experience: 12 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 12 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: VHDL, SPI, DSP, FPGA, ILA, Xilinx Vivado, AXI4, Verilog, Intel Quartus, Python, Hardware Simulation, UART

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | Cedarwave Systems | Jan 2022 - Present

- Reduced FPGA resource utilization by 30% through arithmetic sharing
- Verified 210 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 38% after timing-path optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | Ridgeview Engineering | Jan 2018 - Dec 2021

- Improved maximum operating frequency by 24% after timing-path optimization
- Lowered end-to-end data-path latency by 33% using pipelined logic

Associate FPGA Design Engineer | Urbanaut Engineering | Jul 2014 - Dec 2017

- Reduced FPGA resource utilization by 41% through arithmetic sharing
- Verified 665 protocol corner cases with repeatable simulation suites

PROJECTS

Camera Data Bridge | Verilog, UART, Intel Quartus, Hardware Simulation

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

High-Speed Data Logger | Python, Verilog, FPGA, UART

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Blue River College of Technology, 2014
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Timing Closure Training (2024)